

Thermal desorption spectroscopy (TDS) of Phenol on the Ag(111) surface

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1 Introduction

2 Experimental

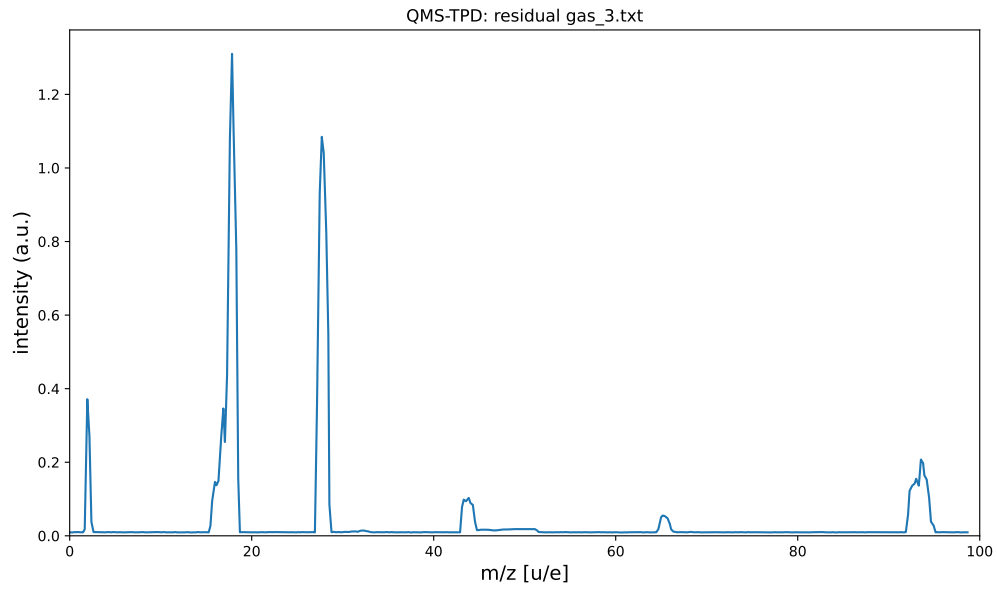


Figure 1: Residual gas analysis of the UHV chamber.

3 Results

3.1 Residual gas analysis

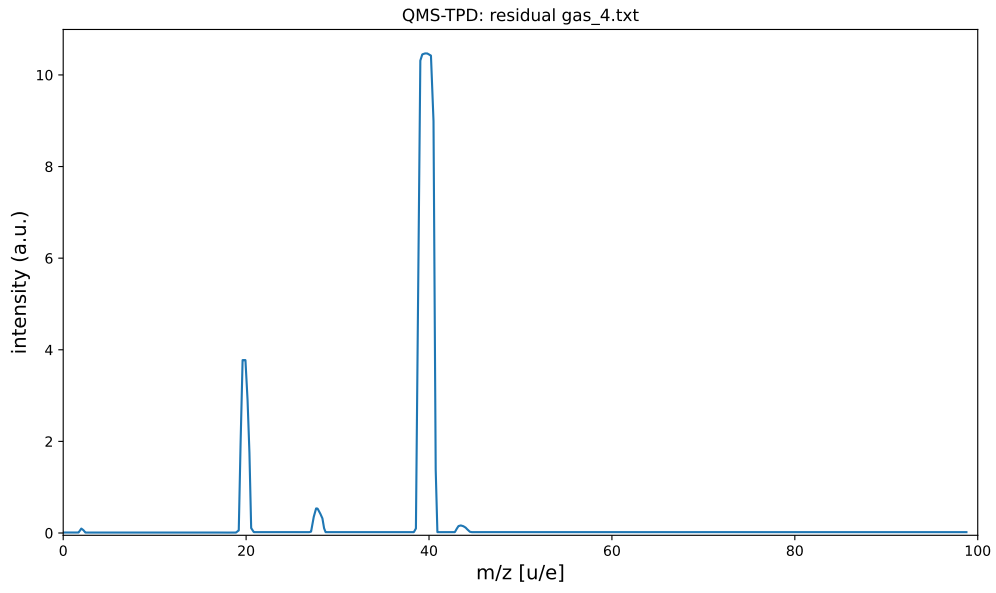


Figure 2: Residual gas analysis of the UHV chamber with argon.

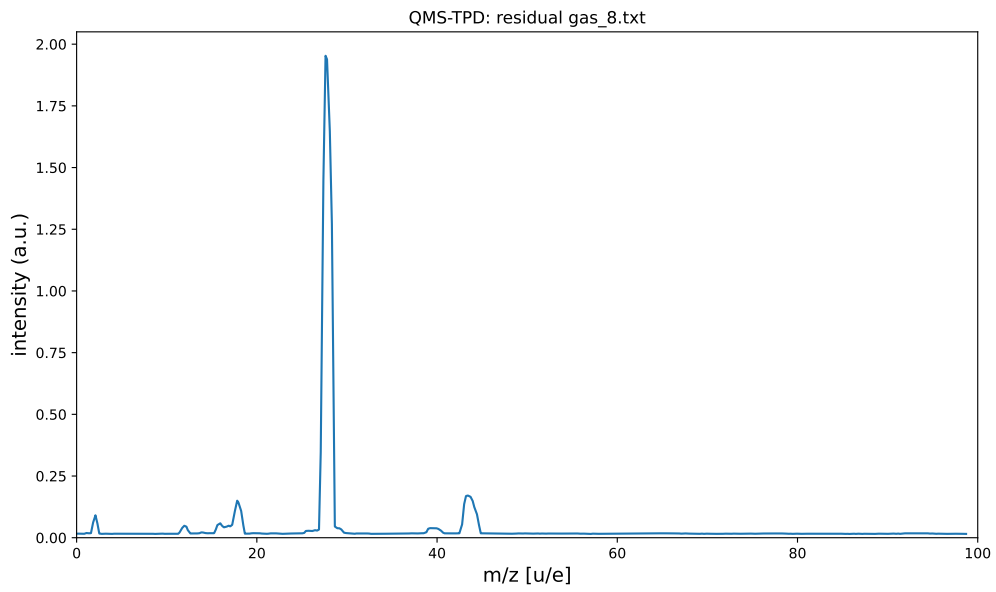


Figure 3: Residual gas analysis of the UHV chamber while degasing the filament.

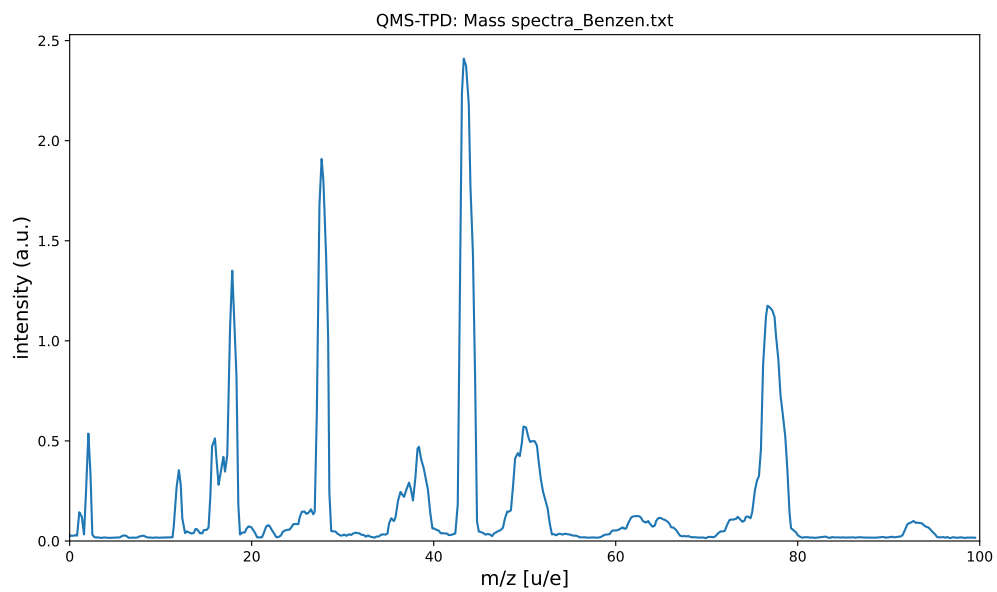


Figure 4: Mass spectrum of benzene.

3.2 Mass Spectra for Phenol and Benzene

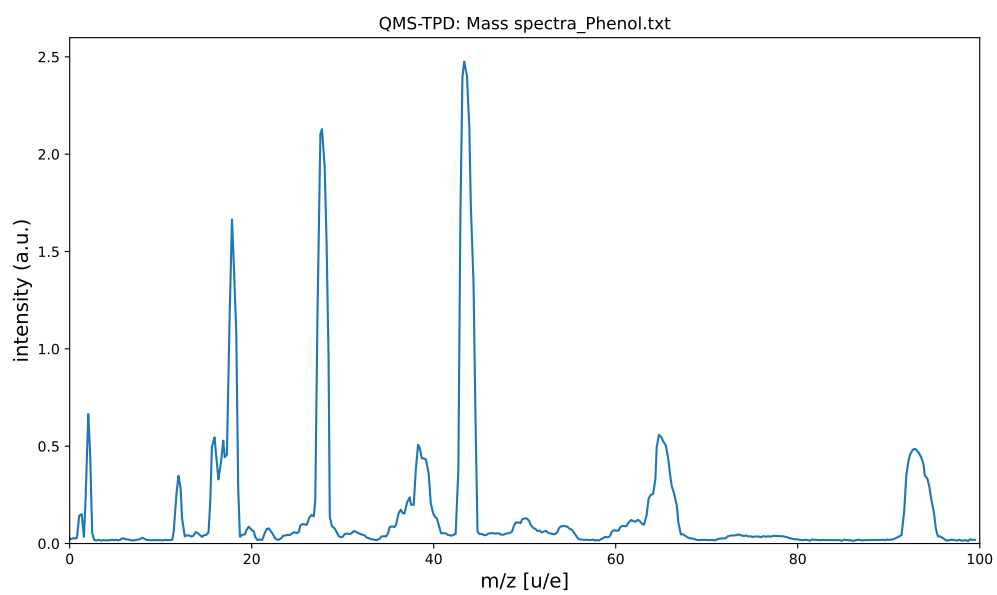


Figure 5: Mass spectrum of phenol.

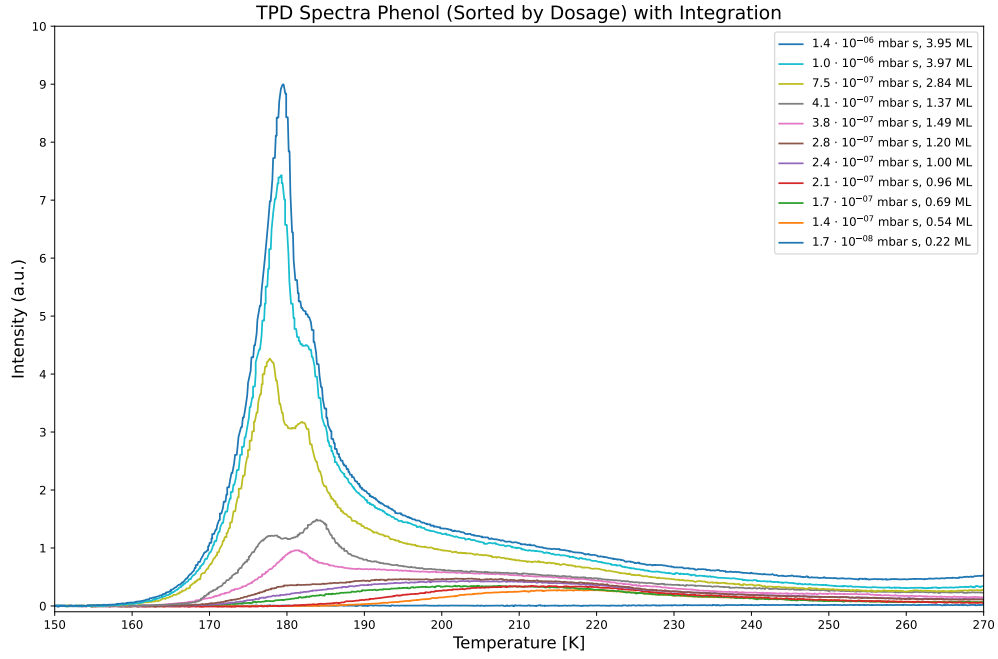


Figure 6: TDS spectrum of phenol on Ag(111) with different coverages.

3.3 TDS Spectra of Phenol on Ag(111)

3.4 Geometric Model of the Monolayer

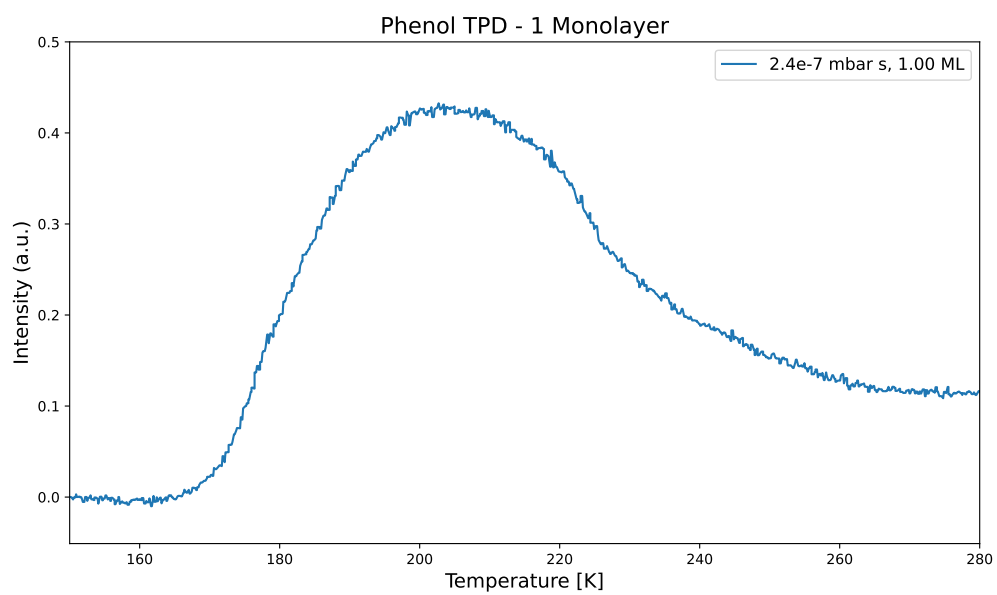


Figure 7: TDS spectrum of 1 **ML!** (**ML!**) of phenol on Ag(111).

Table 1: Dosage, integral of the TDS spectra and coverage for phenol on Ag(111). The coverage is calculated from the TDS spectra.

Dosage [mbar s]	Integral range [K]	Integral [a.u.]	Coverage [ML]
$1.70 \cdot 10^{-8}$	155-270	$9.08 \cdot 10^0$	0.22
$1.40 \cdot 10^{-7}$	155-270	$2.27 \cdot 10^1$	0.54
$1.70 \cdot 10^{-7}$	155-270	$2.88 \cdot 10^1$	0.69
$2.10 \cdot 10^{-7}$	155-270	$4.02 \cdot 10^1$	0.96
$2.40 \cdot 10^{-7}$	155-270	$4.17 \cdot 10^1$	1.00
$2.80 \cdot 10^{-7}$	155-270	$5.02 \cdot 10^1$	1.20
$3.80 \cdot 10^{-7}$	155-268	$6.20 \cdot 10^1$	1.49
$4.10 \cdot 10^{-7}$	155-265	$5.72 \cdot 10^1$	1.37
$7.50 \cdot 10^{-7}$	155-260	$1.19 \cdot 10^2$	2.84
$1.00 \cdot 10^{-6}$	155-258	$1.66 \cdot 10^2$	3.97
$1.40 \cdot 10^{-6}$	155-258	$1.65 \cdot 10^2$	3.95

3.5 Behavior of Coverage and Dosage

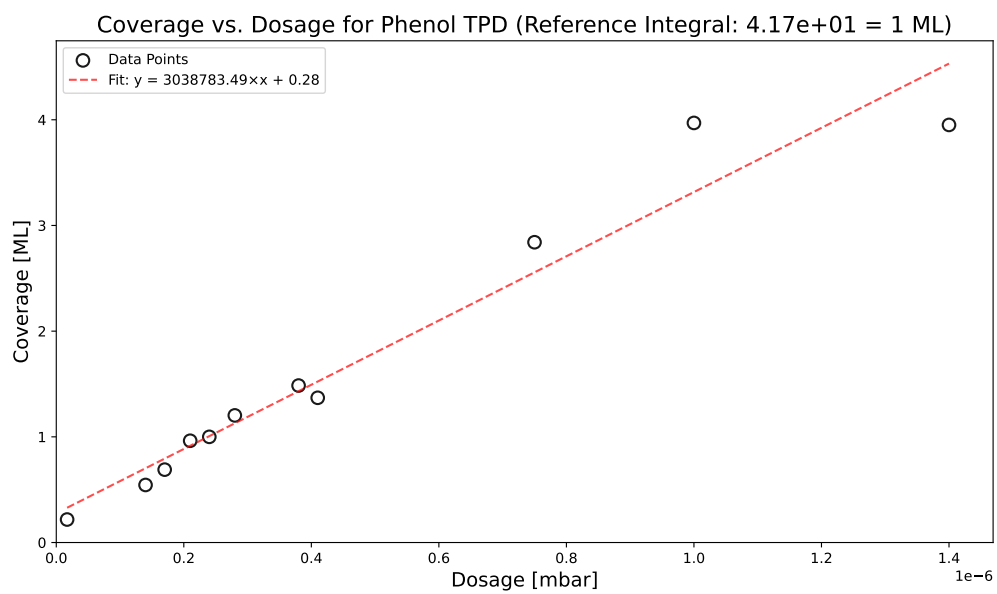
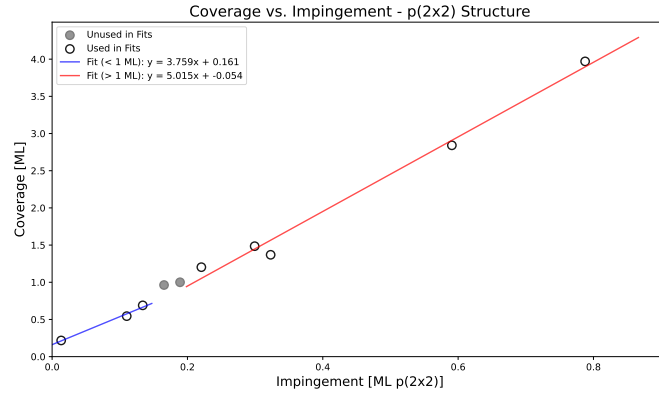


Figure 8: Coverage as a function of dosage for phenol on Ag(111). The coverage is calculated from the TDS spectra.

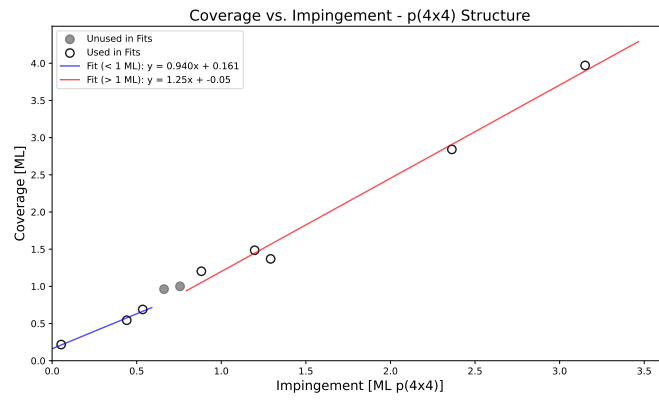
Table 2: Dosage, coverage, and impingement rate for phenol on Ag(111).

Dosage D [mbar s]	Coverage θ [ML]	N_{imp} [cm ⁻²]
$1.70 \cdot 10^{-8}$	0.22	$4.62 \cdot 10^{12}$
$1.40 \cdot 10^{-7}$	0.54	$3.80 \cdot 10^{13}$
$1.70 \cdot 10^{-7}$	0.69	$4.62 \cdot 10^{13}$
$2.10 \cdot 10^{-7}$	0.96	$5.71 \cdot 10^{13}$
$2.40 \cdot 10^{-7}$	1.00	$6.52 \cdot 10^{13}$
$2.80 \cdot 10^{-7}$	1.20	$7.61 \cdot 10^{13}$
$3.80 \cdot 10^{-7}$	1.49	$1.03 \cdot 10^{14}$
$4.10 \cdot 10^{-7}$	1.37	$1.11 \cdot 10^{14}$
$7.50 \cdot 10^{-7}$	2.84	$2.04 \cdot 10^{14}$
$1.00 \cdot 10^{-6}$	3.97	$2.72 \cdot 10^{14}$
$1.40 \cdot 10^{-6}$	3.95	$3.80 \cdot 10^{14}$

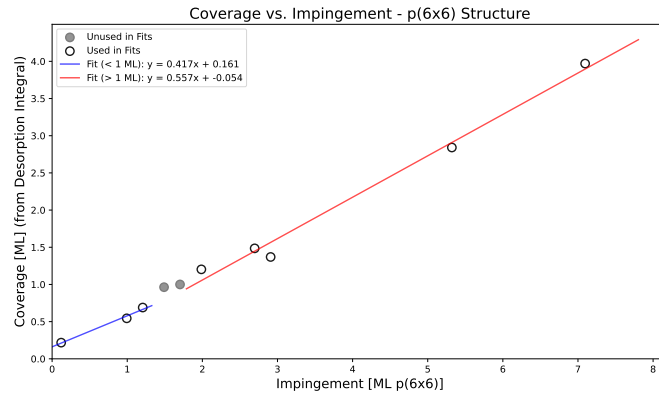
3.6 Sticking Factor



(a) p(2×2) model



(b) p(4×4) model



(c) p(6×6) model

Figure 9: Coverage as a function of impingement rate for phenol on Ag(111) with different geometric models.

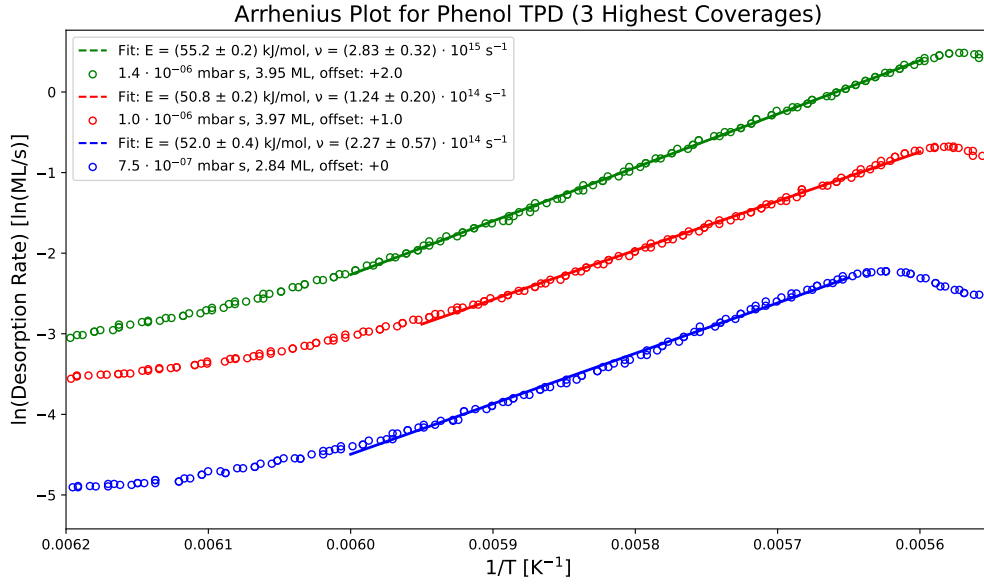


Figure 10: Menzel-Schlichting plot for the multilayer of phenol on Ag(111). The desorption energy is calculated from the slope and the pre-exponential factor is calculated from the intercept of the Menzel-Schlichting plot.

Table 3: Desorption parameters from Menzel-Schlichting analysis for phenol on Ag(111).

Coverage θ [ML]	Desorption Energy E [kJ/mol]	Pre-exponential Factor ν [s ⁻¹]
2.84	52.0 ± 0.4	$(2.27 \pm 0.57) \cdot 10^{14}$
3.97	50.8 ± 0.2	$(1.24 \pm 0.20) \cdot 10^{14}$
3.95	55.2 ± 0.2	$(2.82 \pm 0.32) \cdot 10^{15}$

3.7 Desorption Energy and Pre-Exponential Factor for the Multilayer

3.8 Monolayer Desorption

4 Discussion

5 Appendix

Acronyms

TDS Thermal desorption spectroscopy